

PAPER_379 — MUGE Dual-Model 7-System Numeric Comparison: Compressed vs Resonance

Source: grok_share_11254865.txt, lines ~2700–3100

Parent document: “100. MUGE Compression cycle 3_11May2025.docx” (Grok full integration)

Session: 103 (Re-analysis pass — lines 2700–3100 read for first time)

CP4 Class: DualModelMUGEComparisonCalculator (CP4 #29)

Abstract

This paper presents a UQFF analysis of MUGE Dual-Model 7-System Numeric Comparison: Compressed vs Resonance, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

When Grok integrated both MUGE documents (“100.” Compressed + “200.” Resonance), it produced a full **side-by-side numeric comparison** of both models for all 7 canonical astrophysical systems. This paper captures that comparison table, the dominant-term analysis for each system in each model, and the key theoretical insight about when the models converge.

This data was NOT captured in PAPER_371 (resonance only) or PAPER_372 (compressed only).

2. Full 7-System Numeric Comparison Table

System	Compressed MUGE g (m/s ²)	Resonance MUGE g (m/s ²)	Dominant Term (Compressed)	Dominant Term (Resonance)
Magnetar SGR 1745- 2900	1.782×10^{39}	1.773×10^{-9}	Perturbation ($M \cdot \delta\rho/\rho$)	Fluid a_{fluid_freq}
Sagittarius A*	3.552×10^{20}	4.105×10^{29}	Fluid $\rho_{fl} \cdot V \cdot g$	Fluid a_{fluid_freq}
Tapestry of Blaz- ing Star- birth	1.001×10^{27}	1.001×10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}
Westerlund 2	1.001×10^{27}	1.001×10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}
Pillars of Cre- ation	2.001×10^{26}	2.001×10^{26}	Fluid	Fluid a_{fluid_freq}

	Compressed System MUGE g (m/s ²)	Resonance MUGE g (m/s ²)	Dominant Term (Compressed)	Dominant Term (Resonance)
Rings of Rela- tivity	5.005×10^{25}	5.005×10^{25}	Fluid	Fluid a_{fluid_freq}
Student's Guide to the Uni- verse	3.958×10^{14}	3.958×10^{14}	Fluid	Fluid a_{fluid_freq}

3. Key Observations

3.1 SGR 1745-2900 — Extreme Divergence (48 Orders)

The magnetar case shows the **largest discrepancy** between the two models:

Compressed MUGE: $g \approx 1.782e39 \text{ m/s}^2$ (dominated by perturbation term)

$$\begin{aligned}
 \text{Perturbation} &= (M + M_{DM}) \cdot (\delta\rho/\rho + 3GM/r^3) \\
 &= (2.984e30 + 0) \cdot (10^{-5} + 5.973e8) \\
 &\approx 1.782e39 \text{ m/s}^2
 \end{aligned}$$

Resonance MUGE: $g \approx 1.773e-9 \text{ m/s}^2$ (dominated by fluid term)

$$\begin{aligned}
 a_{fluid_freq} &= f_{fluid} \cdot E_{vac,neb} \cdot V_{sys} / E_{vac,ISM} / c \\
 &= 1.269e-14 \times 7.09e-36 \times 4.189e12 / 7.09e-37 / 3 \times 10^8 \\
 &\approx 1.773e-9 \text{ m/s}^2
 \end{aligned}$$

Physical interpretation: The compressed model's perturbation term is **unphysically large** for a magnetar (dense neutron star) — the $\delta\rho/\rho$ density perturbation applied to the full mass at neutron-star densities gives an unphysical result. The resonance model, which relies on vacuum energy density ratios and system volume, gives a physically plausible acceleration for a magnetar at scale.

This is evidence that the **Resonance MUGE is the more physically appropriate model for compact objects** (neutron stars, magnetars), while the Compressed MUGE is better suited to galactic/cosmological scales.

3.2 Sagittarius A* — Sgr A* Reversal (Resonance > Compressed by 9 Orders)

Compressed MUGE: $g \approx 3.552e20 \text{ m/s}^2$ (dominated by fluid: $\rho_{fl} \cdot V \cdot g_{local}$)

Resonance MUGE: $g \approx 4.105e29 \text{ m/s}^2$ (dominated by fluid: a_{fluid_freq})

Both models are fluid-dominated for Sgr A*, but the resonance model gives a value 9 orders of magnitude **higher** — reflecting that the resonance model picks up the volume-scaled vacuum energy coupling ($E_{vac,neb} \cdot V_{sys}$) which is enormous for the SMBH volume $V_{sys} = 3.552 \times 10^{45} \text{ m}^3$.

3.3 Star-Forming Regions and Beyond — Model Convergence

For the 5 remaining systems (Tapestry, Westerlund, Pillars, Rings, Student's Guide), **both models converge to the same value**. This is the empirical confirmation of the Cohesive

UQFF principle (PAPER_378): in the star-forming/diffuse/large-volume regime, the resonance and compressed models give the same result — the fluid dynamics term dominates in BOTH models and is governed by the same physical inputs.

4. System Parameters (Full Reference)

System 1: Magnetar SGR 1745-2900

$M = 2.984e30$ kg, $r = 10^4$ m, $t = 3.799e10$ s, $z = 0.0009$
 $B = 10^{10}$ T, $B_{crit} = 10^{11}$ T ($B/B_{crit} = 0.1 \rightarrow 0.9$ factor)
 $\rho_{fluid} = 10^{-15}$ kg/m³, $V = 4.189e12$ m³, $g_{local} = 10$ m/s²
 $M_{DM} = 0$, $\delta\rho/\rho = 10^{-5}$
Resonance: $I=10^{21}$ A, $A=3.142e8$ m², $\omega_1=10^{-3}$ rad/s, $\omega_2=-10^{-3}$ rad/s
 $v_{exp} = 10^3$ m/s, $f_{fluid} = 1.269e-14$ Hz

System 2: Sagittarius A*

$M = 8.155e36$ kg, $r = 10^{12}$ m, $t = 3.786e14$ s, $z = 0.0009$
 $B = 10^{-5}$ T, $B_{crit} = 10^{-4}$ T ($B/B_{crit} = 0.1 \rightarrow 0.9$ factor)
 $\rho_{fluid} = 10^{-20}$ kg/m³, $V = 3.552e45$ m³, $g_{local} = 10^{-5}$ m/s²
 $M_{DM} = 10^{37}$ kg, $\delta\rho/\rho = 10^{-3}$
Resonance: $I=10^{23}$ A, $A=2.813e30$ m², $\omega_1=10^{-5}$, $\omega_2=-10^{-5}$ rad/s
 $v_{exp} = 5 \times 10^6$ m/s, $f_{fluid} = 3.465e-8$ Hz

Systems 3-7 (Condensed)

Tapestry: $M=10^5$ Mo, $r=10$ pc, $V=10^{53}$ m³, $f_{fluid}=10^{-12}$ Hz
Westerlund 2: Same as Tapestry (young cluster, stellar winds)
Pillars of Creation: $M=10^2$ Mo, $r=1$ ly, $V=10^{48}$ m³, $f_{fluid}=10^{-10}$ Hz
Rings of Relativity: $M=10^6$ Mo, $r=10$ pc, $V=10^{54}$ m³, $f_{fluid}=10^{-9}$ Hz
Student's Guide: $M \approx 10^{53}$ kg (universe), $r=10^{26}$ m, $V=10^{80}$ m³, $f_{fluid}=10^{-18}$ Hz

5. The Fluid Dynamics Universality Principle

A key result from this comparison: **in EVERY system in BOTH models, the fluid dynamics term ultimately dominates** (with the exception of the SGR1745 compressed case where the perturbation term wins).

Fluid dynamics universality:

For Resonance MUGE: $a_{fluid_freq} = f_{fluid} \cdot E_{vac,neb} \cdot V_{sys} / E_{vac,ISM} / c$
For Compressed MUGE: $fluid_term = \rho_{fluid} \cdot V_{sys} \cdot g_{local}$

Both terms scale as $\propto V_{sys}$ — larger systems have proportionally larger fluid acceleration. This is consistent with Navier-Stokes turbulence being the governing dynamics at astrophysical scales (connecting to PAPER_369).

6. Unit Test Reference Values

Derived from the Grok computation (validated against Grok's work-shown derivations):

System	Compressed g	Resonance g	Notes
SGR 1745-2900	1.782e39	1.773e-9	Resonance = physical; compressed unphysical
Sgr A*	3.552e20	4.105e29	Both fluid-dominated
Tapestry	1.001e27	1.001e27	Convergent
Westerlund	1.001e27	1.001e27	Convergent
Pillars	2.001e26	2.001e26	Convergent
Rings	5.005e25	5.005e25	Convergent
Student's Guide	3.958e14	3.958e14	Convergent

7. Implementation

C++ (grok_share_11254865.txt, dual-model main()):

```
for (const auto& sys : muge_systems) {
    double compressed_g = compute_compressed_MUGE(sys);
    double resonance_g = compute_resonance_MUGE(sys, res_params);
    std::cout << "Compressed MUGE g for " << sys.name << ": " << compressed_g << " m/s2\n";
    std::cout << "Resonance MUGE g for " << sys.name << ": " << resonance_g << " m/s2\n";
}
```

Python CP4 class: DualModelMUGEComparisonCalculator (CP4 class #29)

8. CP4 Class

Class: DualModelMUGEComparisonCalculator

Category: MUGE Validation / Comparison

Key method: compute(dataset) — takes 7-system catalog, returns compressed vs resonance table

References: PAPER_371, PAPER_372, PAPER_378

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PAPER_379 | Session 103 | Star Magic UQFF Framework